Manga109Script: Manga Script Dataset for Script2Layout -supplementary material-

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A Training Conditions

Model training was performed using the `transformers` and `peft` libraries with the following configuration:

- **Dataset:**
  - The `Manga109ScriptDataset` is used with:
    - Maximum token length: 4096,
    - The data sequence is divided into 32 parts.
- **LoRA Configuration:**
  - Reduction factor $r = 8$,
  - `lora_alpha` = 16,
  - Target modules: `q_proj` and `v_proj`,
  - LoRA dropout: 0.1,
  - Bias setting: `none`,
  - Task type: `CAUSAL_LM`.
- **Training Parameters:**
  - **Batch Size:** 1 per device for both training and evaluation,
  - **Gradient Accumulation:** 32 steps,
  - **Epochs:** 20,
  - **Optimizer:** `adamw_torch_fused`,
  - **Learning Rate:** 2×10^{-4},
  - **LR Scheduler:** Cosine scheduler,
  - **FP16:** Enabled,
  - **Logging:** Every 100 steps,
  - **Evaluation Strategy:** Evaluation performed every 100 steps with data grouped by length,

The training is managed through the `Trainer` class provided by the `transformers` library.[1] Finally, after training, the model is saved along with the LoRA adapter parameters (saved separately) and the tokenizer.

B Additional Qualitative Results

In addition to Fig. 3 of the main paper, we introduce a failure case of script2layout models in Fig. B.1. An extreme case is the excessive enlargement of speech balloons. This occurs when there is a large amount of dialogue, and the Script2Layout model directly takes the dialogue content as input.

[1] https://pypi.org/project/transformers/ (cited 2025-04-11)

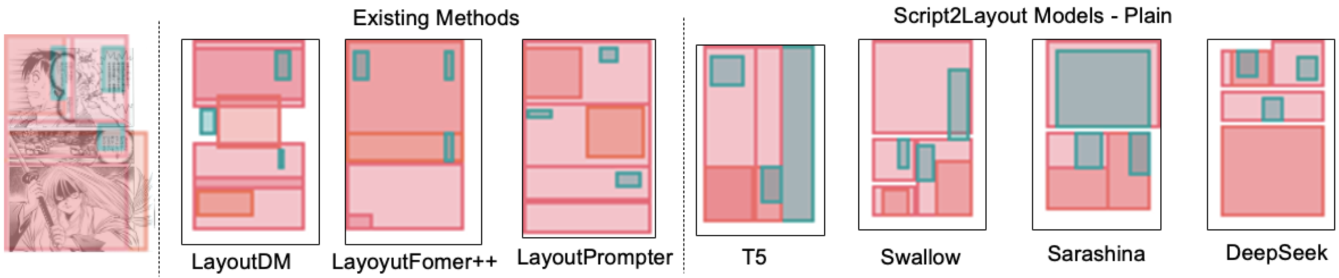

Figure B.1: Failure cases of generated layouts. Each row shows one example: the real manga layout (left), followed by predictions from existing methods and Script2Layout models as in Fig. 3 of the main paper (Source: *Touta Mairimasu* by Shinji Saijo)